

MACHENG, China

WMO: 573990

Lat: 31.1309N

Lon: 114.9547E

Elev: 247

StdP: 14.57

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	28.9	8.0	8.4	37.6	13.1	10.8	38.3	16.3	43.5	14.5	40.9	4.1	0	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	97.1	80.3	95.3	79.9	93.4	79.1	82.7	92.2	81.8	91.2	80.9	90.1	7.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	160.2	87.2	79.4	153.9	86.7	78.5	149.1	86.3	47.0	92.4	45.9	91.2	45.0	90.6	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.0	12.9	11.3	DB	22.1	100.3	2.9	1.9	20.0	101.6	18.3	102.7	16.6	103.7	14.5	105.1	
			WB	19.8	84.2	2.5	1.3	18.0	85.1	16.5	85.9	15.1	86.6	13.3	87.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	40.0	44.6	53.6	64.1	72.9	79.2	84.7	83.6	76.4	66.1	54.4	43.7
	DBStd	16.49	5.52	6.93	7.28	6.84	5.26	4.60	4.63	5.00	5.55	5.98	7.02	5.52
	HDD50	777	315	177	43	2	0	0	0	0	0	0	34	206
	HDD65	2848	776	571	359	95	5	0	0	0	1	60	321	660
	CDD50	5779	4	26	156	425	709	875	1075	1042	791	498	168	10
	CDD65	2376	0	0	7	68	249	425	610	577	342	93	5	0
	CDH74	25021	0	3	57	526	1990	4179	7794	6908	2954	577	33	0
	CDH80	11109	0	0	9	123	677	1666	3983	3361	1157	131	2	0
Wind	WSAvg	4.9	4.6	4.8	5.0	5.3	5.0	4.6	5.3	5.3	5.1	4.6	4.6	4.5
Precipitation	PrecAvg	47.9	1.1	2.0	3.2	5.0	5.9	7.5	8.5	4.7	3.5	3.3	2.2	0.8
	PrecMax	73.9	2.4	5.9	6.9	10.4	13.9	19.5	33.4	15.1	9.2	13.3	6.1	3.4
	PrecMin	26.9	0.0	0.0	0.4	0.8	1.3	0.2	0.9	0.7	0.2	0.1	0.0	0.0
	PrecStd	11.4	0.6	1.4	1.5	2.4	2.9	4.4	6.6	3.9	2.3	3.0	1.6	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.0	71.4	80.0	88.0	92.8	95.9	99.5	99.8	95.7	88.2	78.1	64.7
		MCWB	49.0	57.8	65.4	68.6	72.1	76.1	80.9	80.6	77.7	69.5	60.7	50.9
	2%	DB	56.4	64.6	74.2	83.6	89.7	93.2	97.4	97.2	92.7	83.6	73.7	59.7
		MCWB	46.3	52.4	60.2	66.9	71.1	76.9	80.9	80.5	76.4	67.3	59.3	48.6
	5%	DB	52.6	60.2	70.3	80.1	86.9	90.8	95.6	95.2	89.6	80.4	69.4	56.1
		MCWB	44.2	49.8	58.0	65.4	70.5	76.3	80.6	80.5	74.3	65.0	57.0	46.6
	10%	DB	49.4	55.9	66.2	76.4	83.8	88.4	93.7	93.0	86.5	76.9	65.6	52.9
		MCWB	42.4	48.2	55.9	63.3	69.5	75.8	80.3	79.7	73.0	63.8	55.4	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	60.1	67.9	73.2	77.2	81.8	84.2	84.2	81.0	74.7	65.4	54.0
		MCDB	57.7	69.0	77.1	82.7	86.0	90.0	93.0	92.9	90.9	83.0	72.8	59.5
	2%	WB	48.4	55.1	63.1	70.2	75.2	80.3	82.9	82.7	79.5	70.6	61.8	50.8
		MCDB	53.5	60.4	71.4	78.7	82.8	88.5	92.5	92.0	88.5	78.0	68.7	56.4
	5%	WB	45.6	51.7	60.0	67.9	73.6	79.0	82.0	81.8	78.1	68.3	59.5	48.5
		MCDB	51.4	57.4	67.9	75.8	81.9	86.6	91.7	91.1	86.0	75.8	66.0	54.2
	10%	WB	43.1	49.2	57.1	65.7	71.9	77.9	81.3	80.8	76.2	66.5	57.2	46.3
		MCDB	48.1	54.8	63.6	73.0	79.6	85.0	90.7	89.9	83.4	73.7	64.5	51.5
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.0	15.6	16.4	15.7	13.6	12.8	13.9	15.3	15.9	16.2	15.0
		MCDBR	20.9	22.4	22.3	22.0	21.1	17.6	15.8	16.7	19.3	22.2	22.7	22.2
		MCWBR	12.0	12.2	10.8	9.5	7.2	5.7	4.2	4.5	5.9	8.0	10.1	11.9
	5% WB	MCDBR	18.0	17.9	18.8	17.6	15.3	13.6	14.7	15.2	15.3	16.7	17.5	18.4
		MCWBR	11.5	11.0	10.4	9.4	6.9	6.0	4.9	5.0	6.3	7.7	9.1	10.8
Clear-Sky Solar Irradiance	taub		0.619	0.676	0.708	0.705	0.687	0.714	0.664	0.688	0.687	0.673	0.632	0.603
	taud		1.589	1.525	1.496	1.548	1.616	1.601	1.732	1.664	1.631	1.585	1.618	1.632
	Ebn at Noon		182	188	196	206	211	205	214	207	199	186	178	177
	Edh at Noon		71	82	90	89	83	84	74	78	78	76	68	65
All-Sky Solar Radiation	RadAvg		670	761	1109	1325	1414	1399	1532	1420	1222	987	798	665
	RadStd		106	187	89	140	135	137	170	162	147	137	104	121

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page